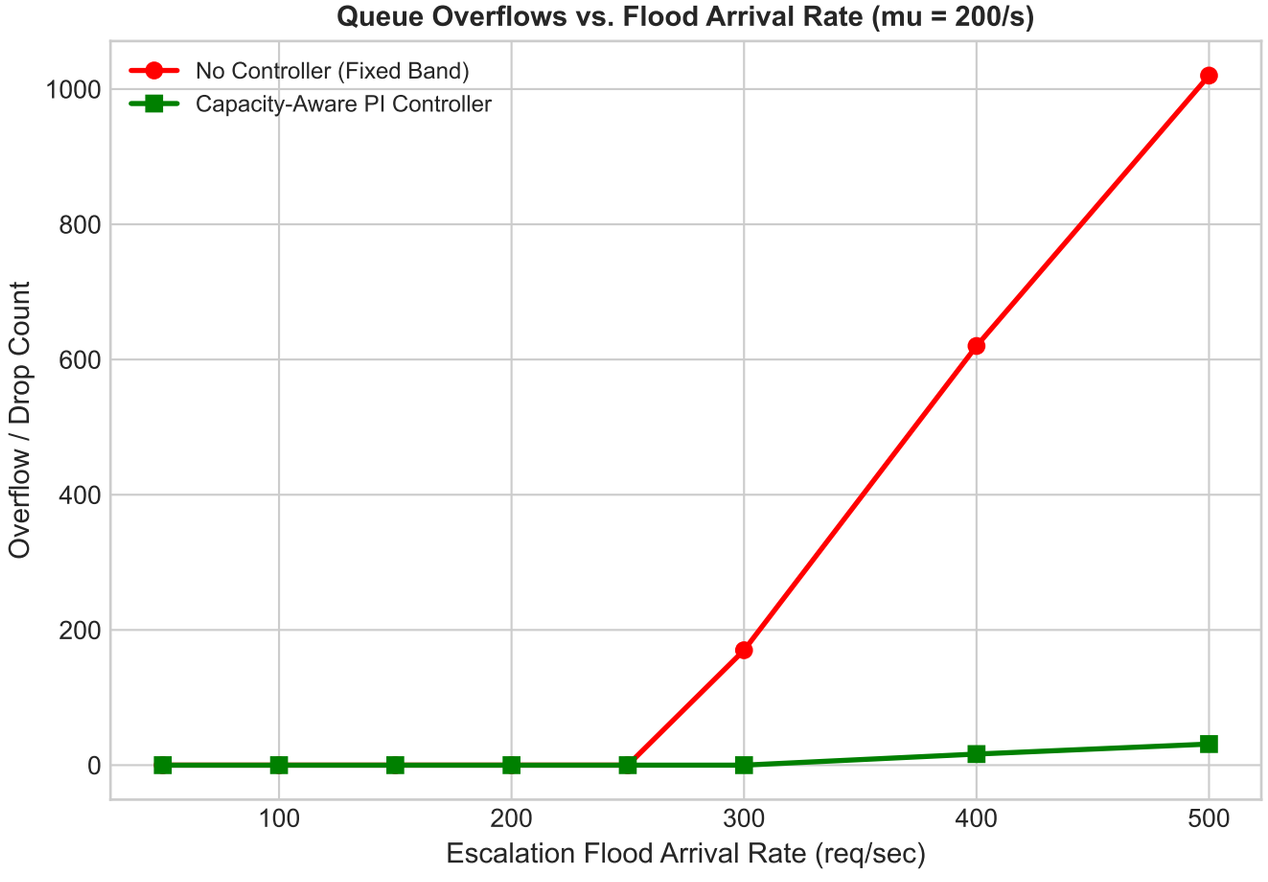
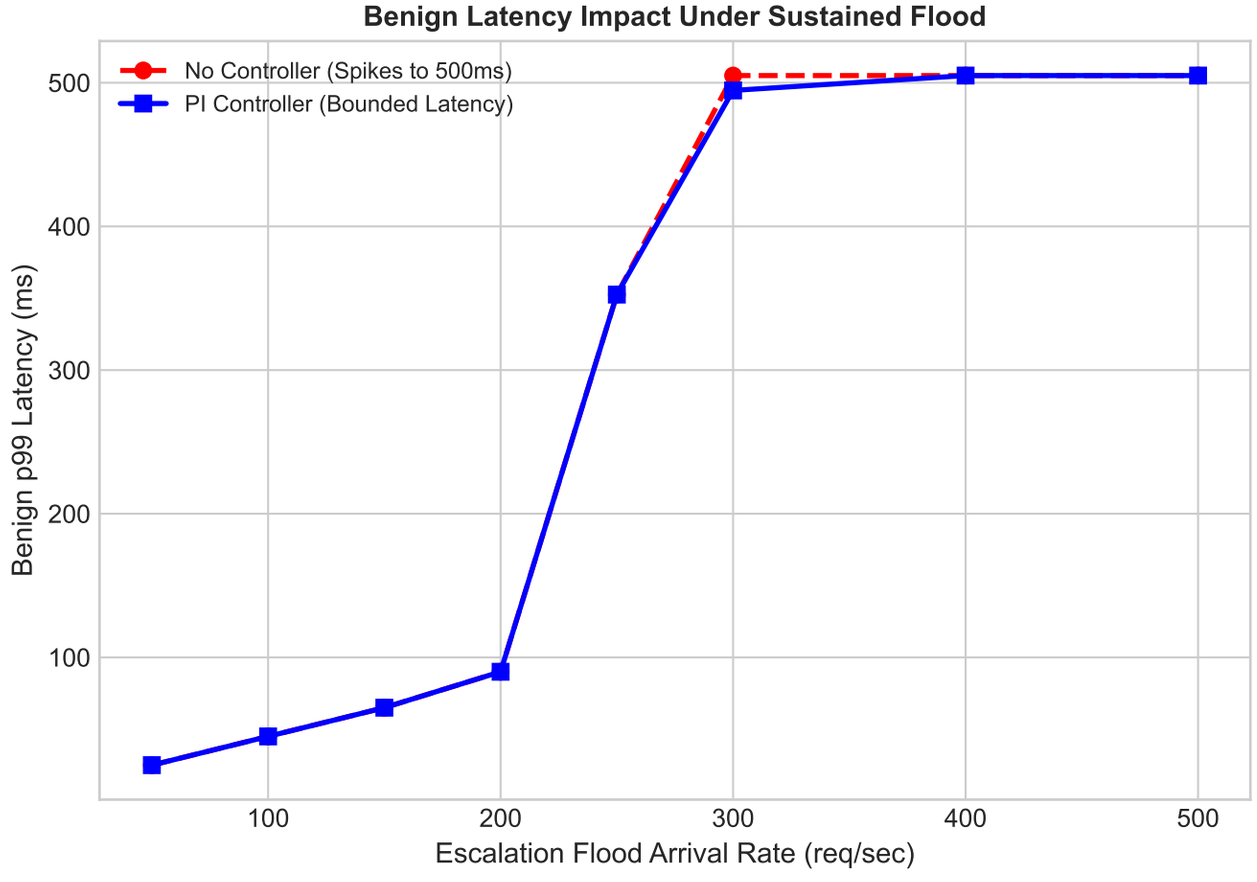

Queue Overflows vs. Flood Arrival Rate (mu = 200/s)
No Controller (Fixed Band)
Capacity-Aware PI Controller
Overflow / Drop Count
0
200
400
600
800
1000
100
200
300
400
500
Escalation Flood Arrival Rate (req/sec)


Benign Latency Impact Under Sustained Flood
No Controller (Spikes to 500ms)
PI Controller (Bounded Latency)
Benign p99 Latency (ms)
100
200
300
400
500
100
200
300
400
500
Escalation Flood Arrival Rate (req/sec)